

Feedback-Augmented RT-DETR

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Same-checkpoint ablation, COCO val2017 @ 640

33.26_{OFF} → 34.25_{ON} → +0.99 _{ΔAP_S}

The story

The data

1. Small objects are hard for transformer detectors.

On COCO val2017, RT-DETR-R50 reports $AP_S = 34.7$ vs $AP_L = 67.7$.
Encoder memory is computed once. The decoder cannot revise it — refinement is one-way.

2. Let the decoder feed back.

After decoder layer 1 makes preliminary predictions, refine the encoder memory with a single cross-attention pass:

$$\mathbf{m}' = \mathbf{m} + g \cdot \text{Attn}(\mathbf{m}, \mathbf{t}^{(1)})$$

Layers 2..L then attend over the refined memory.

3. v1 trained — but contributed nothing.

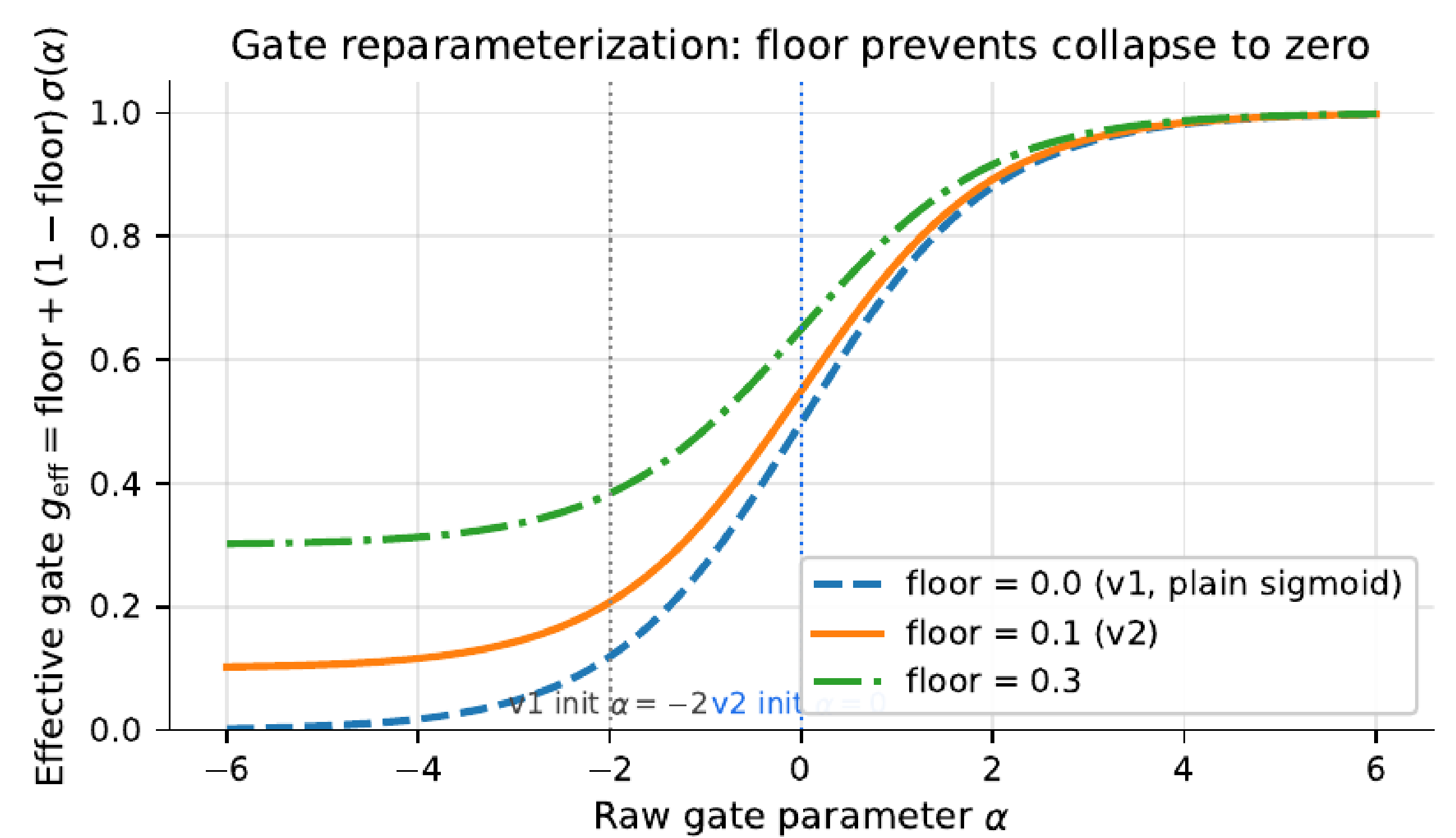
A plain sigmoid gate $g = \sigma(\alpha)$ decayed to ≈ 0.12 during training. The cross-attention output was multiplied by zero before reaching memory:

$$\Delta AP_S = 0.00$$

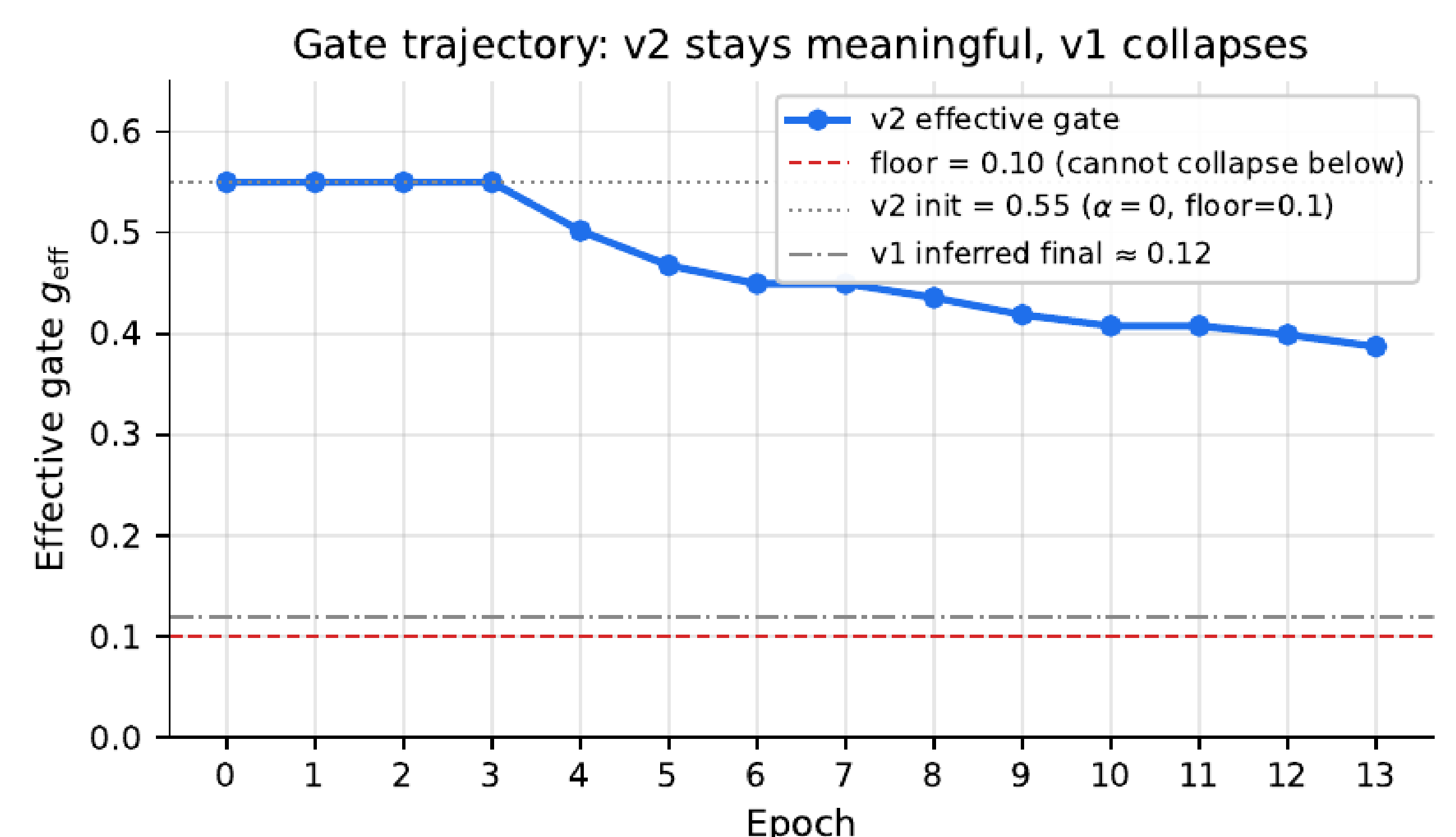
4. Fix it structurally — zero new params.

(i) **Gate floor.** $g_{\text{eff}} = 0.1 + 0.9 \sigma(\alpha)$. The optimizer can attenuate, never silence.

(ii) **Level mask: P2/P3 only.** Refine stride-4 and stride-8 features — where small objects live. Leave P4, P5 untouched.



Floor reparameterization. Plain sigmoid (dashed) saturates to zero; v2 (solid) is bounded below at 0.1 — always non-trivial.



v2 gate stays at ≈ 0.39 over 13 epochs; v1 collapsed to ≈ 0.12 — below v2's floor.

The methodological lesson:

An unconstrained sigmoid gate gives the optimizer a costless route to silence a new module. A floor on the gate is a one-line that closes that route.

